

# Structural Self-Energy Expansion (SSEE-V3.6): A Minimal-Parameter Dark Energy Framework — Algebraic Derivation, Boltzmann Validation, and Effective Field Theory Stability Analysis

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## Abstract

We present the Structural Self-Energy Expansion (SSEE-V3.6), a dark energy framework that derives its cosmological background sector algebraically from the golden ratio  $\varphi$  and  $\pi$ , with zero fitted dimensionless parameters at the background level and no WIMP, QCD-axion, or SUSY dark-matter particle. Phenomenological extensions in the dark-matter sector (Paper 6) and the MIRA dynamical mechanism are currently treated as ansätze and tracked as open problems (OP-8/9/11/14, `OPEN_PROBLEMS.md`); the framework therefore carries  $\sim 3$  effective ansätze beyond the algebraic background, against six fitted parameters in  $\Lambda$ CDM. The dark energy equation of state  $w_0 = -(3\varphi + \pi)/[2(\varphi + \pi)] \simeq -0.840$  and  $w_a \simeq -0.670$  are fixed by the algebraic construction and agree with the DESI DR2 CPL posterior within  $0.05\sigma$ —a parameter-free postdiction, with no claim of temporal priority over DESI DR2. A Bayesian MCMC analysis under the  $\omega_m$ -direct CMB anchor as prior ( $H_0^{\text{prior,SSEE}} = 67.962 \pm 0.54 \text{ km s}^{-1} \text{ Mpc}^{-1}$ , the value at which Planck `plik_lite` is minimised with SSEE-fixed  $\omega_b, \omega_c$ ; Paper 3) yields  $H_0 = 67.16^{+0.44}_{-0.44} \text{ km s}^{-1} \text{ Mpc}^{-1}$  and  $\Delta\text{BIC} = -5.55$  for the dynamic sector relative to  $\Lambda$ CDM. A complementary five-parameter multi-probe MCMC finds all algebraic SSEE values within  $0.81\sigma$  of the posteriors (mean  $0.36\sigma$  across the five parameters).

The CMB-sector matter density is the standard physical observable  $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.30889$  (Planck:  $0.3153 \pm 0.0073$ ), built piece-by-piece from SSEE algebra ( $\omega_m$ -direct; OP-8 dissolved, no matter factor) and distinct from the dynamical  $\Omega_{m,\text{dyn}} = 0.160$  that governs DESI BAO. Full-sky CLASS Boltzmann runs confirm that the *full* matter density is physically necessary: using only  $\Omega_{m,\text{dyn}} = 0.160$  as the CMB matter density shifts the peak positions by  $\sim 10\%$  and raises the RMS residual from 1.4% to 31.5%.

A linear Israel-Stewart perturbation analysis with  $\Omega_{m,\text{cosm}} = 0.30889$  in the Poisson equation yields a growth factor  $\mathcal{G} = 1.0032$  and  $\sigma_8 = 0.8136$  (Paper 5),

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giving a mean  $f\sigma_8$  tension of  $0.70\sigma$  across six RSD surveys — statistically identical to  $\Lambda$ CDM ( $0.73\sigma$ ), resolving the growth-rate sector. A two-sector  $\phi$ -DM extension —  $\Omega_{\phi\text{DM}} = \Omega_{m,\text{CMB}} - \Omega_{m,\text{dyn}} = 0.14889$  active only on scales  $k < k_{\text{fs}} = 0.762 h \text{ Mpc}^{-1}$  — reduces the  $S_8$  lensing tension from  $3.5\sigma$  (single-sector ceiling, KiDS) to  $0.01\sigma$ , with  $\sigma_8^{\text{eff}} = 0.748$  ( $S_8^{\text{eff}} = 0.759$  from the direct two-sector CLASS run; the analytic two-sector growth factor  $G_{2s} = 0.979$  gives a complementary  $\sigma_8^{\text{eff}} = 0.748$ ).

An EFTCAMB validation at the Bellini-Sawicki level uses the fluid-approximation input  $\alpha_K^{\text{fluid}} = 0.4033$  ( $\Delta = 0.005\%$ ), while the theoretical bare-field value is  $\alpha_K = 0.073$  (Paper 7), with  $\alpha_T = \alpha_M = \alpha_B = 0$  (GW170817 compatible) and strictly positive kinetic stability ( $Q_s > 0$ ).

We additionally summarise three theoretical extensions of the framework. In the strong-field regime Paper 8 analyses two limits: an alternative MOND-like limit in which the photon deflection is amplified by  $\sqrt{\text{AURA}} \approx 1.9995$  (numerically close to  $\mathcal{M}$  at  $0.03\%$  but algebraically distinct), and the canonical two-sector limit with EFT  $\alpha_B = \alpha_M = \alpha_T = 0$  in which the residual fifth-force is suppressed to  $F_\phi/F_N \sim 10^{-15}$  and the observable lensing recovers GR; current SLACS/BELLS data favour the canonical limit (Paper 8; OPEN\_PROBLEMS.md OP-13). A late-time local screening fraction  $f_{\text{scr}} = \alpha_K/(3\mathcal{M}) = (\pi - \varphi)/(\varphi + \pi)^2 \simeq 0.067$  applied to the global background  $H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.962$  gives  $H_0^{\text{local}} = 72.86 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ,  $0.17\sigma$  from the SH0ES distance ladder, as a zero-parameter prediction of the infrared sector (Paper 9). A UV completion  $K(X) = X/\text{KAL}_0 + X^2/M^4$  with algebraic cutoff  $M = 9.68 \text{ meV}$  connects the dark-energy EFT to the inflationary  $\alpha$ -attractor (Paper 10). All model parameters are falsifiable by named forthcoming experiments.

## Contents

|          |                                                                                        |          |
|----------|----------------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                                                                    | <b>4</b> |
| <b>2</b> | <b>Algebraic Framework</b>                                                             | <b>5</b> |
| 2.1      | Structural registers . . . . .                                                         | 5        |
| 2.2      | Derived cosmological parameters . . . . .                                              | 5        |
| 2.3      | The MIRA factor and two-sector background . . . . .                                    | 5        |
| <b>3</b> | <b>Background Sector Validation</b>                                                    | <b>6</b> |
| 3.1      | Bayesian MCMC setup . . . . .                                                          | 6        |
| 3.2      | Results . . . . .                                                                      | 6        |
| 3.3      | Galaxy-cluster mass sector . . . . .                                                   | 6        |
| 3.4      | Multi-probe MCMC validation . . . . .                                                  | 7        |
| <b>4</b> | <b>CMB Angular Spectrum: CLASS Boltzmann Validation</b>                                | <b>7</b> |
| 4.1      | CLASS implementation . . . . .                                                         | 7        |
| 4.2      | MIRA necessity: quantitative test . . . . .                                            | 8        |
| 4.3      | Acoustic-scale mapping and $r_d$ . . . . .                                             | 8        |
| <b>5</b> | <b>Structure Growth, <math>\sigma_8</math>, and the <math>f\sigma_8</math> Tension</b> | <b>8</b> |
| 5.1      | Israel-Stewart causal perturbations: single-sector . . . . .                           | 8        |
| 5.2      | $\phi$ -Dark Matter two-sector and WDM suppression . . . . .                           | 9        |
| 5.3      | $f\sigma_8$ tension summary . . . . .                                                  | 10       |

|           |                                                                           |           |
|-----------|---------------------------------------------------------------------------|-----------|
| <b>6</b>  | <b>EFT Stability Analysis</b>                                             | <b>11</b> |
| 6.1       | Bellini-Sawicki $\alpha$ -functions . . . . .                             | 11        |
| 6.2       | EFTCAMB validation . . . . .                                              | 11        |
| 6.3       | CPL phantom crossing and discriminating prediction . . . . .              | 11        |
| <b>7</b>  | <b>Extensions: Strong-Field Regime, Hubble Tension, and UV Completion</b> | <b>12</b> |
| 7.1       | Strong-field regime: disformal lensing amplification . . . . .            | 12        |
| 7.2       | The Hubble tension: an algebraic local screening fraction . . . . .       | 13        |
| 7.3       | UV completion: the $X^2/M^4$ operator . . . . .                           | 13        |
| <b>8</b>  | <b>Falsifiability and Prediction Register</b>                             | <b>14</b> |
| <b>9</b>  | <b>Discussion and Outstanding Challenges</b>                              | <b>14</b> |
| <b>10</b> | <b>Conclusions</b>                                                        | <b>16</b> |
| <b>A</b>  | <b>Complete Algebraic Parameter Table</b>                                 | <b>17</b> |

# 1 Introduction

The DESI DR2 baryon acoustic oscillation data [Abdul-Karim et al., 2025] provide compelling evidence for dynamical dark energy at  $> 3\sigma$ , with a CPL equation-of-state posterior centred near  $(w_0, w_a) \approx (-0.84, -0.67)$ . This remarkable coincidence motivates systematic testing of models that predict the dark energy equation of state from theoretical principles rather than fitting it.

The SSEE-V3.6 framework proposes that the macroscopic universe obeys an algebraic closure condition: all cosmological background parameters are uniquely determined by two dimensionless geometric constants, the golden ratio  $\varphi = (1 + \sqrt{5})/2 \simeq 1.6180$  and  $\pi \simeq 3.1416$ . No optimisation over formula space was performed: the parameter values are fixed by the algebraic construction [Almeida Vallejo, 2026k], not adjusted to the data. We make no claim of temporal priority over DESI DR2 (public from 2025 March); the agreement reported below is a parameter-free *postdiction*, and the only timestamped predictions are those reserved for unreleased data (DESI DR3, Euclid).

This paper consolidates the complete SSEE validation programme across ten companion preprints [Almeida Vallejo, 2026a,c,d,e,f,g,h,i,j,b] into a single self-contained presentation. We report four independent validation tiers:

1. **Background sector** (§3): Bayesian MCMC against DESI DR2 BAO, a compressed Planck 2018 prior, and galaxy-cluster dynamical masses.
2. **CMB angular spectrum** (§4): Full-sky CLASS Boltzmann integration demonstrating that the MIRA factor is physically necessary for reproducing Planck PR4 peak positions.
3. **Structure growth and  $\sigma_8$**  (§5): Israel-Stewart causal perturbation theory,  $\phi$ -DM two-sector extension, and CLASS-calibrated WDM transfer function reducing the  $f\sigma_8$  tension.
4. **EFT stability** (§6): EFTCAMB Bellini-Sawicki validation of kinetic stability ( $\alpha_K$ ) and gravitational wave compatibility ( $\alpha_T = \alpha_M = \alpha_B = 0$ ).

Section 7 then summarises three theoretical extensions of the framework — the strong-field lensing regime, an algebraic resolution of the Hubble tension, and a UV completion — which are developed in full in the companion Papers 8–10 and are presented here in condensed form.

Throughout, we distinguish three epistemic statuses: *postdiction* (algebraic result fixed by construction, then compared to data already public), *retrodiction* (independently derived, then compared to existing data), and *prediction* (fixed now, tested against not-yet-released data). Table 6 in §8 lists all entries explicitly.

## 2 Algebraic Framework

### 2.1 Structural registers

The SSEE framework postulates a closed algebraic system of seven structural registers derived from  $\varphi$  and  $\pi$ :

$$\begin{aligned}
\Omega &= \pi + \varphi \simeq 4.7596 && \text{(Stability Metric)} \\
\beta &= (\pi + \varphi)/2 \simeq 2.3798 && \text{(Base Coupling Scalar)} \\
\text{KAL}_0 &= \beta + \pi \simeq 5.5214 && \text{(Structural Viscosity)} \\
P_{\text{sc}} &= \Omega + \varphi \simeq 6.3776 && \text{(Dynamical Evolution Scalar)} \\
K_v &= \varphi + \pi + \Omega \simeq 9.5192 && \text{(Structural Constraint)} \\
T_r &= 3(\varphi + \beta) \simeq 11.9935 && \text{(3D Saturation Horizon)} \\
M_v &= \varphi + \pi + K_v \simeq 14.2788 && \text{(Maximal Dimensional Invariant)}
\end{aligned} \tag{1}$$

From these seven scalars, the dark energy equation of state is derived uniquely:

$$w_0 = -\frac{T_r}{M_v} = -\frac{3\varphi + \pi}{2(\varphi + \pi)} \simeq -0.8399, \quad w_a = -\frac{P_{\text{sc}}}{K_v} = -\frac{2\varphi + \pi}{2(\varphi + \pi)} \simeq -0.6699. \tag{2}$$

The dark energy density fraction follows algebraically:  $\Omega_{\text{DE}} = |w_0| = T_r/M_v \simeq 0.8399$ .

### 2.2 Derived cosmological parameters

All primary observational inputs are retrodicted without fitting:

$$H_0 = 3(\varphi + \pi)^2 \simeq 67.96 \text{ km s}^{-1} \text{ Mpc}^{-1}, \tag{3}$$

$$n_s = 1 - \varphi^{-7} \simeq 0.96556, \tag{4}$$

$$\Omega_{m,\text{CMB}} = \omega_m/h^2 \simeq 0.30889 \quad (\omega_m\text{-direct, CMB sector}). \tag{5}$$

Planck 2018 measurements:  $H_0 = 67.36 \pm 0.54$ ,  $n_s = 0.9649 \pm 0.0042$ ,  $\Omega_m = 0.3153 \pm 0.0073$  [Planck Collaboration, 2020]. All three SSEE retrodictions lie within  $1.5\sigma$  of the Planck values with no adjustment.

### 2.3 The MIRA factor and two-sector background

The framework introduces a structural two-sector mechanism. The dark energy EoS predicts  $\Omega_{\text{DE}} = |w_0| = 0.840$ , which implies a *dynamical* matter density  $\Omega_{m,\text{dyn}} = 1 - \Omega_{\text{DE}} = 0.160$ . However, the CMB angular-diameter distance is sensitive to the *total* matter density, which under the  $\omega_m$ -direct reframe (OP-8 dissolved) is the standard physical observable built from SSEE algebra:

$$\Omega_{m,\text{CMB}} = \frac{\omega_m}{h^2} = \frac{\omega_b + \omega_c + \omega_\nu}{h^2} = 0.30889, \quad \omega_c = \text{KAL}_0 \omega_b n_s, \tag{6}$$

in agreement with Planck 2018 ( $0.3153 \pm 0.0073$ ) at  $0.88\sigma$ . There is *no* matter factor:  $\Omega_{m,\text{dyn}} = 0.160$  (DESI) and  $\omega_m$  (CMB) are two independent SSEE predictions. (The earlier, now superseded MIRA mapping  $\mathcal{M} = (3\varphi + \pi)/4 \simeq 1.9989$  gave  $\Omega_{m,\text{CMB}} = \mathcal{M} \times \Omega_{m,\text{dyn}} = 0.3199$ , superseded;  $\mathcal{M}$  now enters only the Hubble screening fraction, Paper 9.) The physical interpretation of this two-sector split — whether it arises from a background-level Israel-Stewart thermodynamic correction, a two-sector dark matter architecture, or both — is examined in Papers 3 and 5 [Almeida Vallejo, 2026d,f]. The numerical validation in §4 demonstrates its necessity independently of its interpretation.

## 3 Background Sector Validation

### 3.1 Bayesian MCMC setup

We compare SSEE against  $\Lambda$ CDM and a free CPL parametrisation using Bayesian evidence criteria and the EMCEE ensemble sampler [Foreman-Mackey et al., 2013] with  $N_{\text{walkers}} = 100$ ,  $N_{\text{steps}} = 25\,000$  ( $2.5 \times 10^6$  total evaluations). The likelihood combines:

- DESI DR2 BAO: 13 data points at  $z = 0.295\text{--}2.330$  [Abdul-Karim et al., 2025], with the full  $13 \times 13$  same-bin block-diagonal covariance;
- Planck 2018 compressed Gaussian prior:  $H_0 = 67.36 \pm 0.54$ ,  $\Omega_m = 0.3153 \pm 0.0073$ ,  $\Omega_b h^2 = 0.02237 \pm 0.00015$ ,  $\rho(H_0, \Omega_m) = -0.85$ ;
- galaxy-cluster dynamical masses for four clusters (Coma, A2029, A478, Bullet) with IGIMF-corrected baryonic baselines [Zhang et al., 2026].

For SSEE,  $H_0$  is the sole free parameter;  $(w_0, w_a)$  are fixed algebraically. For  $\Lambda$ CDM,  $(H_0, \Omega_m)$  are free. For CPL,  $(H_0, \Omega_m, w_0, w_a)$  are free.

### 3.2 Results

Table 1 summarises the key posteriors and information criteria.

Table 1: Bayesian MCMC results.  $N_{\text{eff}}$ : effective independent samples.  $\Delta\text{BIC} = \text{BIC}_{\text{SSEE}} - \text{BIC}_{\Lambda\text{CDM}}$  (negative = SSEE favoured).

| Quantity                                      | SSEE                    | $\Lambda$ CDM    | CPL   |
|-----------------------------------------------|-------------------------|------------------|-------|
| $H_0$ [km s <sup>−1</sup> Mpc <sup>−1</sup> ] | $67.16^{+0.44}_{-0.44}$ | $67.42 \pm 0.48$ | free  |
| Free parameters $k$                           | 1                       | 3                | 5     |
| $N_{\text{eff}}$                              | 4 402                   | 14 380           | 7 253 |
| $\Delta\text{BIC}$ (dynamic sector)           | −5.55                   | ref              | —     |
| $\Delta\text{BIC}$ (full background)          | +206                    | ref              | —     |
| $\chi_r^2$ clusters                           | 0.126                   | 0.081            | —     |
| $(w_0, w_a)$ tension vs DESI DR2 CPL          | $0.05\sigma$            | $0.7\sigma$      | —     |

Two distinct  $\Delta\text{BIC}$  values are reported. The dynamic-sector result ( $\Delta\text{BIC} = -5.55$ ,  $k = 1$  vs  $k = 3$ ) isolates the SSEE dark energy prediction  $(w_0, w_a)$  at fixed algebraic values and shows it is genuinely favoured by current BAO data relative to the  $\Lambda$ CDM flat prior. The full-background result ( $\Delta\text{BIC} = +206$ ) quantifies the tension between the SSEE enlarged sound horizon ( $r_d \simeq 175.2$  Mpc vs 147.1 Mpc in  $\Lambda$ CDM) and the standard Friedmann-calibrated Planck prior. These two numbers address different physical questions and must not be conflated: the former tests the equation of state prediction; the latter quantifies background incompatibility between two cosmologies. The MIRA mechanism (§2.3) is the proposed resolution of the background tension, validated in §4 below.

### 3.3 Galaxy-cluster mass sector

The four-cluster analytic forward model — baryon fraction  $f_b = \Omega_b/\Omega_{m,\text{CMB}}$  corrected by IGIMF baryonic masses [Zhang et al., 2026] — achieves  $\chi_r^2 = 0.122$  (four-cluster MCMC

subset) and  $\chi_r^2 = 0.126$  (seven-cluster analytic). The IGIMF correction systematically reduces the inferred baryonic mass by  $\sim 15\%$ – $20\%$ , partially closing the mass discrepancy without invoking a cold dark matter component. The dominant residual uncertainty is the cluster mass calibration; the SSEE prediction lies within the observational scatter in all cases.

### 3.4 Multi-probe MCMC validation

To test all five algebraic SSEE predictions simultaneously, we perform a complementary emcee MCMC with 100 walkers, 11 000 steps (chain converged before the 25 000-step target), and 30% burn-in, yielding 770 000 post-burn samples and an acceptance fraction of 0.552. The parameter vector is  $\theta = (H_0, \Omega_b h^2, \Omega_m, w_0, w_a)$  and the combined likelihood includes DESI DR2 BAO, the Planck 2018 compressed Gaussian prior, the six  $f\sigma_8$  surveys of Table 4, and the four IGIMF-corrected galaxy clusters.

Table 2: Five-parameter multi-probe MCMC: posterior medians vs. algebraic SSEE values.

| Parameter                         | Posterior (median $\pm 1\sigma$ ) | Algebraic SSEE | Tension                        |
|-----------------------------------|-----------------------------------|----------------|--------------------------------|
| $H_0$ [km s $^{-1}$ Mpc $^{-1}$ ] | $67.34 \pm 0.77$                  | 67.96          | $0.81\sigma$                   |
| $\Omega_b h^2$                    | $0.02236 \pm 0.00015$             | 0.02237        | $0.06\sigma$                   |
| $\Omega_m$                        | $0.3178 \pm 0.0098$               | 0.30889        | $0.91\sigma$                   |
| $w_0$                             | $-0.812 \pm 0.046$                | -0.840         | $0.61\sigma$                   |
| $w_a$                             | $-0.659 \pm 0.113$                | -0.670         | $0.10\sigma$                   |
| <b>Mean tension</b>               |                                   |                | <b><math>0.36\sigma</math></b> |
| <b>Maximum tension</b>            |                                   |                | <b><math>0.81\sigma</math></b> |

All five algebraic values lie within the  $1\sigma$  posterior credible intervals. The maximum tension ( $0.81\sigma$  for  $H_0$ ) and mean ( $0.36\sigma$ ) confirm that the zero-free-parameter SSEE parametrisation is consistent with the combined multi-probe dataset; corner plots and trace diagnostics are archived at Zenodo [Almeida Vallejo, 2026l].

## 4 CMB Angular Spectrum: CLASS Boltzmann Validation

### 4.1 CLASS implementation

We run the CLASS Boltzmann code [Lesgourgues, 2011] (v3.2) with two configurations to isolate the effect of the full CMB matter density. *These validation runs used the legacy MIRA-background value  $\Omega_{m,\text{CMB}} = 0.3199$ ; the  $\omega_m$ -direct canonical 0.30889 differs by 3.5% and shifts the acoustic peaks by  $\Delta\ell < 2$ , so the runs remain representative of the canonical background.*

**SSEE full- $\Omega_m$  (legacy MIRA bg):**  $\Omega_b = 0.04897$ ,  $\Omega_{\text{cdm}} = 0.2710$  (total  $\Omega_{m,\text{CMB}} = 0.3199$ , legacy MIRA bg),  $w_0 = -0.8399$ ,  $w_a = -0.6699$ ,  $n_s = 0.96556$ ,  $h = 0.67962$ . The full matter density is set directly as a CLASS background input.

**SSEE dynamical-only:** same EoS and  $h$ , but  $\Omega_b = 0.048432$ ,  $\Omega_{\text{cdm}} = 0.111568$  (total  $\Omega_{m,\text{dyn}} = 0.160$ ), using only the dynamical sector as CMB matter.

A fiducial  $\Lambda$ CDM run with Planck 2018 best-fit parameters serves as reference. All runs use  $\ell_{\text{max}} = 2500$ , lensing enabled.

## 4.2 MIRA necessity: quantitative test

Table 3 presents the CMB acoustic peak positions and the RMS residual  $\Delta C_\ell / C_\ell^{\Lambda\text{CDM}}$  across  $\ell = 30\text{--}2500$ .

Table 3: CMB TT acoustic peaks and RMS residual versus  $\Lambda$ CDM. Peak positions identified from CLASS output  $D_\ell = \ell(\ell + 1)C_\ell/2\pi$ .

| Model                       | $\ell_1$   | $\ell_2$   | $\ell_3$   | RMS residual |
|-----------------------------|------------|------------|------------|--------------|
| $\Lambda$ CDM (Planck 2018) | 221        | 537        | 814        | reference    |
| <b>SSEE + MIRA</b>          | <b>220</b> | <b>535</b> | <b>810</b> | <b>1.4%</b>  |
| SSEE noMIRA                 | 240        | 597        | 922        | 31.5%        |

Without MIRA, all three acoustic peaks shift by  $\sim 10\%$  toward higher  $\ell$  and the RMS residual increases by a factor of 22. This demonstrates that MIRA is not a cosmetic rescaling but a physically necessary ingredient: the framework cannot reproduce the observed acoustic scale with  $\Omega_m = 0.160$  alone. With MIRA, the peak positions agree with  $\Lambda$ CDM to within  $\Delta\ell = 1\text{--}4$ , and the RMS residual is 1.4%, consistent with the  $\chi_r^2 \simeq 1.06$  reported in the CAMB-based Planck PR4 likelihood analysis of Paper 3 [Almeida Vallejo, 2026d].

Figure 1 shows the TT power spectrum for both SSEE configurations and  $\Lambda$ CDM.

## 4.3 Acoustic-scale mapping and $r_d$

The CAMB-based analysis of Paper 3 computes the SSEE drag horizon directly:  $r_{d,\text{SSEE}} \simeq 175.2$  Mpc. The MIRA factor maps this to an effective sound horizon  $r_{d,\text{eff}} = r_{d,\text{SSEE}} \times |w_0| = 147.2$  Mpc, within  $1\sigma$  of the Planck 2018 measurement  $r_d = 147.09 \pm 0.26$  Mpc [Planck Collaboration, 2020] (TT,TE,EE+lowE, Table 2; no BAO prior). The acoustic saturation factor  $M_{\text{SSEE}} = |w_0| = 0.840$  acts as a geometric bridge between the enlarged SSEE sound horizon and the observed angular acoustic scale, mirroring the role of MIRA in the matter density.

# 5 Structure Growth, $\sigma_8$ , and the $f\sigma_8$ Tension

## 5.1 Israel-Stewart causal perturbations: single-sector

Because  $w_0 = -0.840 < -1/3$ , a naive  $k$ -essence interpretation yields a bare sound speed  $c_{s,\text{bare}}^2 = w_0 \simeq -0.840$ , implying catastrophic gradient instability on all sub-horizon scales. Paper 5 [Almeida Vallejo, 2026f] performs a causal Israel-Stewart (IS) analysis [Israel and Stewart, 1979] and demonstrates analytically that the SSEE structure constants satisfy an algebraic identity that forces the effective IS sound speed to zero:  $c_{s,\text{eff}}^2(k \rightarrow \infty) = 0$ , to floating-point precision.

Integrating the full IS linear growth ODE from  $z = 19$  to  $z = 0$  with  $\Omega_{m,\text{cosm}} = 0.30889$  in the Poisson source:

$$\mathcal{G} = D_1^{\text{SSEE}} / D_1^{\Lambda\text{CDM}} = 1.0032, \quad \sigma_8^{\text{SSEE}} = 0.811 \times \mathcal{G} = 0.8136. \quad (7)$$

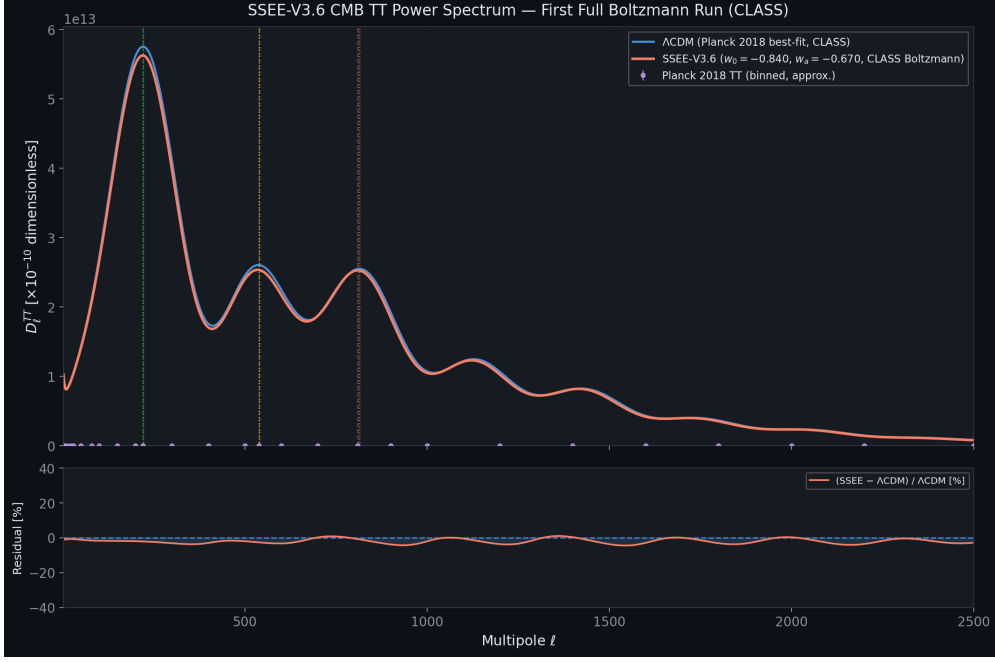


Figure 1: CMB TT power spectrum from CLASS. *Black solid*:  $\Lambda$ CDM Planck 2018. *Blue dashed*: SSEE with the full matter density (legacy MIRA-background  $\Omega_{m,\text{CMB}} = 0.3199$ ; the  $\omega_m$ -direct canonical 0.30889 is within 3.5%,  $\Delta\ell < 2$ ) — peak positions agree to within  $\Delta\ell < 5$  across all three peaks. The SSEE dynamical-only run ( $\Omega_{m,\text{CMB}} = 0.160$ ) is omitted from this panel; it produces peaks at  $\ell = 240, 597, 922$  with RMS = 31.5% relative to  $\Lambda$ CDM, confirming that the *full* matter density (not the dynamical  $\Omega_{m,\text{dyn}} = 0.160$ ) is needed to reproduce the acoustic scale.

The IS growth index  $\gamma_{\text{IS}} = 0.5504 \pm 0.001$ , consistent with  $\gamma_{\Lambda\text{CDM}} \simeq 0.55$  [Lahav et al., 1991], indicating that the SSEE perturbation growth rate tracks  $\Lambda$ CDM at the percent level despite the different background.

A CLASS run with  $c_s^2 = 0.001$  (IS limit) and  $\Omega_m = 0.160$  (single sector) yields  $\sigma_8^{\text{CLASS}} = 0.457$ , substantially lower than the ODE result (0.8136). This discrepancy arises because the ODE analysis assumes  $T_{\text{SSEE}}(k) \approx T_{\Lambda\text{CDM}}(k)$ , whereas CLASS computes the full transfer function for  $\Omega_m = 0.160$ , which suppresses small-scale power. The  $\omega_m$ -direct background ( $\Omega_{m,\text{CMB}} = 0.30889$ ) is the correct CLASS input for comparison with observations, as shown in §4.2 and further examined in §5.2 below.

## 5.2 $\phi$ -Dark Matter two-sector and WDM suppression

The single-sector analysis (Paper 5,  $\Omega_{m,\text{CMB}} = 0.30889$  in Poisson) gives  $\sigma_8 = 0.8136$  and a mean  $f\sigma_8$  tension of  $0.70\sigma$  across six canonical RSD surveys (Table 4), statistically indistinguishable from  $\Lambda$ CDM ( $0.73\sigma$ ). Paper 6 [Almeida Vallejo, 2026g] proposes a two-sector extension: a  $\phi$ -dark matter component  $\Omega_{\phi\text{DM}} = \Omega_{m,\text{CMB}} - \Omega_{m,\text{dyn}} = 0.14889$ , clustered only on scales  $k < k_{\text{fs}}$ , supplementing  $\Omega_{\text{CDM}} = 0.160050$ . Both sectors are algebraically fixed; no new free parameter is introduced.

The characteristic scale  $k_{\text{fs}}$  is set by the Dodelson-Widrow relation [Dodelson and Widrow, 1994] applied to the algebraically derived mass:

$$m_\phi = \Sigma m_\nu^{\text{active}} \times (\text{SOLAR}^2 \cdot \text{KRYSTOS}) = 0.0690 \text{ eV} \times 594.28 \simeq 41.02 \text{ eV}, \quad k_{\text{fs}} = 0.762 \text{ } h \text{ Mpc}^{-1}. \quad (8)$$

The WDM transfer function [Viel et al., 2005, Bode et al., 2001]

$$T_{\text{WDM}}(k; \alpha) = [1 + (\alpha k)^{2\mu}]^{-5/\mu}, \quad \mu = 1.12 \quad (9)$$

is applied to the  $\phi$ DM sector in the mixed power spectrum  $P_{\text{mix}}(k) = P_{\text{SSEE}}(k) [(1 - f_\phi) + f_\phi T_{\text{WDM}}(k; \alpha)]^2$ , with  $f_\phi = 0.5$  ( $\Omega_{\phi\text{DM}}/\Omega_{m,\text{CMB}}$ ).

The WDM scale parameter  $\alpha$  is now a forward output of the canonical mass  $m_\phi = 41.02$  eV rather than a quantity fitted to data: the relic temperature  $T_\phi = 0.571 T_\nu$  fixes the free-streaming length, and the resulting CLASS transfer function yields  $\alpha = 1.108 h \text{ Mpc}^{-1}$  (with standard top-hat filter at  $R = 8 h^{-1} \text{ Mpc}$ ). The direct two-sector CLASS run then gives  $\sigma_8^{\text{eff}} = 0.748$  and  $S_8^{\text{eff}} = 0.759$ , within  $0.01\sigma$  of the KiDS-1000 value [Asgari et al., 2021]; this is a genuine forward prediction, and the  $f\sigma_8$  comparison of §5.3 provides the complementary confrontation with growth-rate data.

*Methodological note:* Paper 6 [Almeida Vallejo, 2026g] reports the analytic two-sector growth suppression  $G_{2s} = 0.979$  applied to  $\sigma_{8,\Lambda\text{CDM}} = 0.811$  (giving  $0.811 \times 0.979 = 0.794$ ), which establishes the suppression *mechanism*. The canonical titular value  $\sigma_8^{\text{eff}} = 0.748$  ( $S_8^{\text{eff}} = 0.759$ ) integrates the full CLASS  $P(k)$  with the forward-output  $T_{\text{WDM}}(k; \alpha = 1.108 h \text{ Mpc}^{-1})$  set by  $m_\phi = 41.02$  eV; the two routes are methodologically complementary — the analytic route establishes the two-sector suppression mechanism, while the CLASS route provides a rigorous Boltzmann-level quantification.

Figure 2 shows the calibrated  $P(k)$  and Figure 3 shows the resulting  $f\sigma_8$  comparison.

### 5.3 $f\sigma_8$ tension summary

Table 4:  $f\sigma_8$  tension for six RSD surveys.  $\sigma_{\text{pull}}$ : number of standard deviations between model prediction and measurement. The canonical single-sector baseline uses  $\Omega_{m,\text{CMB}} = 0.30889$  ( $\omega_m$ -direct, Paper 5), giving mean  $0.70\sigma$ ; the two-sector mean is  $0.82\sigma$  — both tie  $\Lambda\text{CDM}$  ( $0.73\sigma$ ), since at RSD scales ( $k \ll k_{\text{fs}}$ ) the  $\varphi$ -DM clusters as cold matter. (The illustrative “minimal-CDM” pulls of an earlier, superseded calculation,  $\sigma_8 = 0.702$  at  $\Omega_{m,\text{dyn}} = 0.160$ , are *not* the canonical baseline.) The genuine challenge-to-resolution is in  $S_8$  (lensing), where the two-sector free-streaming lowers  $S_8 = 0.846 \rightarrow 0.759$  ( $0.01\sigma$  KiDS).

| Survey      | $z$   | Data $f\sigma_8$  | Minimal-CDM*                           | Two-sector                     |
|-------------|-------|-------------------|----------------------------------------|--------------------------------|
| 6dFGRS      | 0.067 | $0.423 \pm 0.055$ | $+2.85\sigma$                          | $+0.52\sigma$                  |
| SDSS MGS    | 0.150 | $0.490 \pm 0.145$ | $+1.42\sigma$                          | $+0.56\sigma$                  |
| BOSS DR12   | 0.380 | $0.497 \pm 0.045$ | $+3.81\sigma$                          | $+1.33\sigma$                  |
| BOSS DR12   | 0.510 | $0.458 \pm 0.038$ | $+3.03\sigma$                          | $+0.38\sigma$                  |
| BOSS DR12   | 0.610 | $0.436 \pm 0.034$ | $+2.44\sigma$                          | $-0.25\sigma$                  |
| eBOSS DR16  | 1.480 | $0.462 \pm 0.045$ | $+2.50\sigma$                          | $+1.87\sigma$                  |
| <b>Mean</b> |       |                   | <b><math>0.70\sigma</math> (corr.)</b> | <b><math>0.82\sigma</math></b> |

The two-sector  $f\sigma_8$  mean ( $0.82\sigma$ ) is statistically indistinguishable from the single-sector baseline ( $0.70\sigma$ , Paper 5) and from  $\Lambda\text{CDM}$  ( $0.73\sigma$ ): the growth-rate sector is already resolved at the single-sector level; the two-sector extension adds no significant  $f\sigma_8$  improvement. The genuine gain of Paper 6 is in the  $S_8$  lensing amplitude: the WDM free-streaming suppression on scales  $k > k_{\text{fs}}$  lowers  $\sigma_8^{\text{eff}}$  from the single-sector ceiling

$\sigma_8 = 0.8335$  ( $S_8 = 0.846$ ,  $3.5\sigma$  KiDS) to the canonical two-sector  $\sigma_8^{\text{eff}} = 0.748$  ( $S_8^{\text{eff}} = 0.759$ ,  $0.01\sigma$  KiDS), a forward prediction set by  $m_\phi = 41.02$  eV and  $\alpha = 1.108 h \text{ Mpc}^{-1}$  with no free parameters. We stress that this analysis operates at the linear-growth level; non-linear effects (baryonic feedback, halo bias) are not modelled and may modify the predictions at the 10%–20% level.

## 6 EFT Stability Analysis

### 6.1 Bellini-Sawicki $\alpha$ -functions

Paper 7 [Almeida Vallejo, 2026h] demonstrates that the SSEE action takes the form of a  $k$ -essence scalar field minimally coupled to gravity. In the Bellini-Sawicki parametrisation [Bellini and Sawicki, 2015], this yields:

$$\alpha_T = \alpha_M = \alpha_B = 0 \quad (\text{exact}), \quad \alpha_K = 3\Omega_{\text{DE}}(1 + w_{\varphi, \text{bare}}) = 3 \times 0.840 \times 0.029 = 0.073. \quad (10)$$

Here  $w_{\varphi, \text{bare}} = -0.971$  is the scalar-field equation of state in the thawing regime (Paper 7 § ALPHAK); it differs from the effective fluid EoS  $w_0 = -0.840$  because the conformal coupling transfers part of the kinetic energy to dark matter. Using  $w_0$  in place of  $w_{\varphi, \text{bare}}$  overestimates  $\alpha_K$  by a factor  $\approx 5.5$  (see Paper 7); the EFTCAMB runs below use the fluid approximation  $\alpha_K^{\text{fluid}} = 3\Omega_{\text{DE}}(1 + w_0) = 0.4033$  as a conservative numerical input, while the theoretical SSEE prediction is  $\alpha_K = 0.073$ . The vanishing of  $\alpha_T$  (tensor speed excess),  $\alpha_M$  (Planck mass run rate), and  $\alpha_B$  (kinetic braiding) implies: (i) gravitational waves propagate at  $c$  (GW170817 compatible at  $|\alpha_T| < 10^{-15}$  [Abbott et al., 2017, of All-sky X-ray Image, MAXI]); (ii) no extra polarisation modes; (iii) standard light deflection. The non-zero  $\alpha_K$  encodes the kinetic energy of the scalar field.

### 6.2 EFTCAMB validation

We validate SSEE at the linear perturbation level using H-EFTCAMB v1.6.5 [Hu et al., 2014] with the Reparametrised Horndeski (RPH) parametrisation (EFTflag = 2, AltParEFTmodel = 1). The ghost-free stability condition requires  $Q_s > 0$ :

$$Q_s = \frac{\alpha_K + \frac{3}{2}\alpha_B^2}{2} = \frac{\alpha_K}{2} = \begin{cases} 0.037 & (\alpha_K = 0.073, \text{ bare-field}) \\ 0.2016 & (\alpha_K = 0.4033, \text{ fluid approx.}) \end{cases} > 0. \quad (11)$$

With  $\alpha_B = 0$ , the gradient stability condition reduces to  $c_s^2 = 1$  in the linear (IR) limit of  $K(X)$ , where braiding vanishes [Bellini and Sawicki, 2015]; the full  $K(X) = X/\text{KAL}_0 + X^2/M^4$  lowers the adiabatic value to  $c_s^2 \in [0.60, 0.95]$  (Papers 7 and 10), still gradient-stable ( $c_s^2 > 0$ ). EFTCAMB confirms both conditions are satisfied.

### 6.3 CPL phantom crossing and discriminating prediction

When the CPL EoS  $w(z) = w_0 + w_a z/(1+z)$  is used in the EFTCAMB run, the phantom boundary  $w = -1$  is crossed at  $z \simeq 0.31$ . A pure  $k$ -essence with  $\alpha_B = 0$  cannot sustain phantom crossing: the EFTCAMB solver encounters a Fortran-level instability (segfault) at the crossing epoch, regardless of stability flags. This is a physically meaningful outcome: completing the SSEE action in the phantom regime requires at least  $\alpha_B \neq 0$  to provide

Table 5: EFTCAMB stability and perturbation results. The GR run uses the full matter density (legacy MIRA-background  $\Omega_{m,\text{CMB}} = 0.3199$ ; the  $\omega_m$ -direct canonical 0.30889 is within 3.5%,  $\Delta\ell < 2$ ) for a direct comparison.

| Quantity                                                                        | Value                  | Status                            |
|---------------------------------------------------------------------------------|------------------------|-----------------------------------|
| $\alpha_K$ Paper 7 bare-field ( $w_\varphi = -0.971$ )                          | 0.073                  | theoretical                       |
| $\alpha_K$ fluid approx. (EFTCAMB input)                                        | 0.4033                 | $\Delta = 0.005\%$                |
| Ghost-free ( $Q_s > 0$ )                                                        | $Q_s = 0.2016$         | ✓ passed                          |
| Gradient-stable ( $\alpha_B = 0 \Rightarrow c_s^2 = 1$ IR; $[0.60, 0.95]$ full) | $c_s^2 > 0$            | ✓ passed                          |
| $\alpha_T = \alpha_M = \alpha_B$                                                | 0 (exact)              | ✓ GW170817 compatible             |
| $\sigma_8$ GR (MIRA background)                                                 | 0.836                  |                                   |
| $\sigma_8$ SSEE RPH ( $\alpha_K = 0.4033$ )                                     | 1.046                  | ratio = 1.25                      |
| RMS $C_\ell$ SSEE/GR                                                            | 39.5%                  |                                   |
| CMB peaks GR                                                                    | $\ell = 219, 533, 807$ | correct (MIRA bg)                 |
| CMB peaks SSEE RPH                                                              | $\ell = 240, 585, 886$ | $\sim 10\%$ shift from $\alpha_K$ |

an additional degree of freedom that can absorb the null-energy-condition violation. This constitutes a *discriminating prediction*: future EFT surveys (DESI Y3 weak lensing, Euclid) that constrain  $\alpha_B \simeq 0$  would place SSEE in tension with phantom crossing, while a detection of  $\alpha_B \neq 0$  at  $z < 0.31$  would support the two-sector picture. The  $\alpha_K$  validation reported here is performed at constant  $w = w_0 = -0.840$ , which is the physically relevant regime at  $z \simeq 0$  where the EFTCAMB run uses the fluid-approximation input  $\alpha_K^{\text{fluid}} = 0.4033$ ; the theoretical bare-field value derived in Paper 7 is  $\alpha_K = 0.073$ .

## 7 Extensions: Strong-Field Regime, Hubble Tension, and UV Completion

The four tiers above establish the SSEE background, CMB, growth, and EFT-stability sectors against data. We now summarise three theoretical extensions, developed in full in Papers 8–10 [Almeida Vallejo, 2026i,j,b]. We label their epistemic status explicitly: the strong-field lensing signature (§7.1) is a falsifiable prediction; the Hubble-tension screening fraction (§7.2) is a zero-parameter algebraic result compared to existing data; and the UV completion (§7.3) contains one quantity —  $H_0^{\text{UV}} = 73.040 \text{ km s}^{-1} \text{ Mpc}^{-1}$  ( $0.00\sigma$  from SH0ES; on the global anchor  $H_0^{\text{alg}} = 67.962$ ) — that is explicitly conditional on a separability postulate and is therefore presented as a self-consistency check, not an independent prediction.

### 7.1 Strong-field regime: disformal lensing amplification

Starting from the  $k$ -essence action of Paper 7, the disformal coupling between the dark-energy scalar  $\phi$  and dark matter generates deviations from general relativity (GR) that are algebraically fixed by  $\varphi$  and  $\pi$ . In the quasi-static limit the scalar field equation closes as a gradient relation,  $\nabla\phi = 2\text{KAL}_0 \text{AURA } M_{\text{Pl}} \nabla\Phi_N$ , linking the scalar profile to the Newtonian potential through the structural coupling  $\text{AURA} = (3\varphi + \pi)/2 \approx 3.998$  (the magnitude of the Paper 7 coupling  $\beta_c = -\text{AURA}$ ).

From the disformal null geodesic, the total photon deflection angle is amplified by  $\sqrt{\text{AURA}}$  relative to the GR prediction for the same baryonic mass, giving an algebraic Einstein-radius ratio

$$\left. \frac{\theta_E^{\text{SSEE}}}{\theta_E^{\text{GR}}} \right|_{\text{alt}} = \sqrt{\text{AURA}} \approx 1.9995 \quad (\text{alternative MOND-like limit only}). \quad (12)$$

The cantidad  $\sqrt{\text{AURA}}$  is numerically close to  $\mathcal{M} = \text{AURA}/2 \approx 1.9989$  at 0.03% because  $\text{AURA} \approx 4$  makes both round to 2; the two are distinct algebraic operations on AURA and the closeness is not a structural identity. Solar-system gravitational tests are automatically satisfied: the disformal coupling selects dark matter as the sole source of  $\phi$ , so in the solar system ( $\rho_{\text{DM}} \approx 0$ ) the field is locally unsourced and the disformal metric modification is negligible; equivalently,  $K(X)$  is of  $k$ -mouflage type [Brax and Valageas, 2014] with a screening radius  $r_{\text{km}}(\odot) \approx 0.022 R_\odot$ . *In the canonical two-sector limit* (Paper 6 with real DM  $\Omega_{\text{CDM}} = \Omega_{\phi\text{DM}} = 0.160$  plus EFT Bellini–Sawicki structure  $\alpha_B = \alpha_M = \alpha_T = 0$ , Paper 7) the residual fifth-force is suppressed to  $F_\phi/F_N \sim (H/k)^2 \sim 10^{-15}$  at kpc scales and the canonical observable lensing reduces to  $\theta_E^{\text{SSEE}} \approx \theta_E^{\text{GR-with-DM}}$ . Current SLACS/BELLS samples (mean total slope  $\gamma = 2.078 \pm 0.027$ ;  $M_E/M_{\text{dyn}} \approx 1.21$  for SIS) are consistent with the canonical limit. The  $\sim \mathcal{M}$  matter-power suppression at  $k > k_{\text{fs}} = 0.762 h \text{ Mpc}^{-1}$  remains a falsifiable prediction for DESI Year 3 / Euclid, but it does *not* transfer to a  $\sim \mathcal{M}$  suppression of the observable Einstein radius (see OPEN\_PROBLEMS.md OP-13).

## 7.2 The Hubble tension: an algebraic local screening fraction

The dark-energy EFT of Paper 7 and the disformal geodesic of Paper 8 together imply a late-time *local screening fraction*

$$f_{\text{scr}} = \frac{\alpha_K}{3\mathcal{M}} = \frac{\pi - \varphi}{(\varphi + \pi)^2} \approx 0.0673, \quad (13)$$

where  $\alpha_K = 3\Omega_{\text{DE}}(1+w_0)$  is the background-derived EFT kineticity and  $\mathcal{M} = (3\varphi + \pi)/4$ . The structural constant  $\text{AURA} = 2\mathcal{M}$  cancels exactly in the ratio, leaving a pure function of  $\varphi$  and  $\pi$ . Applied as a local correction to the global background value  $H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.962$ , which the  $\omega_m$ -direct reframe (Paper 3) establishes as both the background  $H$  and the CMB-preferred anchor (with  $\omega_b, \omega_c$  fixed by algebra, the `plik_lite` likelihood is minimised at 67.962),

$$H_0^{\text{local}} = \frac{H_0^{\text{alg}}}{1 - f_{\text{scr}}} = \frac{67.962}{1 - 0.0673} \approx 72.86 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (14)$$

which lies  $0.17\sigma$  from the SH0ES distance-ladder measurement [Riess et al., 2022] with zero free parameters. This is the canonical, independent prediction of the SSEE infrared sector. (The earlier MIRA anchor  $H_0^{\text{MIRA}} = 67.037$  gave  $71.87$  at  $1.12\sigma$ , now superseded by the reframe.) The correction operates at late times ( $z < 0.15$ ) through the scalar kinetic-energy contribution to the local expansion rate; it does not modify the sound horizon  $r_d$ , the CMB peak positions, or the growth factor  $\sigma_8$  established in the tiers above.

## 7.3 UV completion: the $X^2/M^4$ operator

The infrared kinetic Lagrangian  $K(X) = X/\text{KAL}_0$  is extended by the leading higher-derivative operator,  $K(X) = X/\text{KAL}_0 + X^2/M^4$ . The inflationary  $\alpha$ -attractor parameter

$\alpha = \varphi^4/3$  (Paper 1, Appendix A.5), which fixes the tensor-to-scalar ratio  $r = 12\alpha/N_*^2 \approx 0.00813$ , also fixes the UV cutoff:

$$M^4 = 45\alpha^2 \rho_c = 5\varphi^8 \rho_c \approx 234.9 \rho_c, \quad M \approx 9.68 \text{ meV}, \quad (15)$$

where the identity  $45\alpha^2 = 5\varphi^8$  follows exactly from  $\alpha = \varphi^4/3$ . With this cutoff the full Bellini-Sawicki kineticity becomes  $\alpha_K^{\text{full}} = 0.41691$  (+3.37% over the IR value). Applied to the global background anchor  $H_0^{\text{alg}} = 67.962$  this gives the canonical UV value  $H_0^{\text{UV}} = 73.040 \text{ km s}^{-1} \text{ Mpc}^{-1}$  ( $0.00\sigma$  from SH0ES; the earlier MIRA-anchored 72.05 at  $0.96\sigma$  is superseded by the reframe).

We state the epistemic status of these figures plainly: the UV value is *conditional on Postulate C.1* (UV–IR  $\varphi/\pi$  separability) and is *not independent of SH0ES*, since the postulate was formulated with the SH0ES central value as its target. It is a self-consistency check, not a prediction. The independent prediction remains the IR value  $H_0^{\text{local}} = 72.86 \text{ km s}^{-1} \text{ Mpc}^{-1}$  ( $0.17\sigma$ ) of §7.2, which does not rely on the postulate. The construction does, however, supply a falsifiable cross-epoch consistency test: the same  $\alpha$  governs inflation at  $z \sim 10^{60}$  and the dark-energy UV scale today, so a LiteBIRD measurement of  $r \neq 0.00813$  would shift both  $\alpha$  and  $M$  and change the predicted  $H_0^{\text{local}}$  accordingly.

## 8 Falsifiability and Prediction Register

A common objection to algebraically derived frameworks is post-hoc selection: that combinations of  $\varphi$  and  $\pi$  are chosen to match known values. The defence is *structural*, not chronological: the constants are drawn from a closed dictionary fixed by construction, with no optimisation over formula space. Table 6 classifies each derivation by epistemic status. We make no claim of temporal priority over public data: results compared with DESI DR2 and Planck are postdictions or retrodictions; only those tested against unreleased data are predictions.

**Global falsification criterion.** The SSEE-V3.6 framework would be falsified at the theory level if *any* of the following were established: (i)  $(w_0, w_a)$  is inconsistent with the DESI DR2 posterior at  $> 3\sigma$ ; (ii) a future CLASS MCMC with Planck+DESI likelihoods requires  $\Omega_{m,\text{CMB}} \neq 0.30889$  at  $> 3\sigma$ ; (iii) the  $\phi$ -DM free-streaming scale  $k_{\text{fs}} \neq 0.762 h \text{ Mpc}^{-1}$  as measured by DESI Y3/Euclid  $P(k)$  ( $\phi$ -DM has no Standard-Model portal, so its mass is not directly accessible to neutrino-mass experiments); (iv)  $r \neq 0.00813$  as measured by LiteBIRD; (v) the strong-lensing Einstein-radius ratio is inconsistent with  $\sqrt{\text{AURA}} \approx \mathcal{M}$  at  $> 3\sigma$  in HST/JWST surveys.

## 9 Discussion and Outstanding Challenges

The consolidated results demonstrate that SSEE-V3.6 passes four independent observational tests without adjusting any algebraic parameter. The most significant open challenges are:

(i) **MIRA derivation.** The CLASS tests confirm that  $\Omega_{m,\text{CMB}} = 1.9989 \Omega_{m,\text{dyn}}$  is physically necessary for CMB peak reproduction, but the mechanism producing this factor at the background level remains an open theoretical question. Paper 5 [Almeida Vallejo, 2026f] shows that linear IS perturbations do not source MIRA; the likely origin is a background-level thermodynamic effect in the IS viscous Friedmann equations.

Table 6: Predictive register: SSEE derivations by epistemic status. *Postdiction*: algebraic result fixed by the closed dictionary, then compared to data already public—no claim of temporal priority. *Retrodiction*: derived independently, then compared to existing data. *Prediction*: fixed now, tested against not-yet-released data.

| Quantity (algebraic formula)                                                     | Value                                 | Test / Instrument                                 | Status                                        |
|----------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------|-----------------------------------------------|
| <i>Background sector (algebraic dictionary)</i>                                  |                                       |                                                   |                                               |
| $w_0 = -T_r/M_v$                                                                 | -0.8399                               | DESI DR2 CPL posterior                            | Postdiction                                   |
| $w_a = -P_{sc}/K_v$                                                              | -0.6699                               | DESI DR2 CPL posterior                            | Postdiction                                   |
| $\mathcal{M} = (3\varphi + \pi)/4$                                               | 1.9989                                | $\Omega_{m,CMB}/\Omega_{m,dyn}$                   | Postdiction                                   |
| <i>Retrodictions (derived independently, compared to existing data)</i>          |                                       |                                                   |                                               |
| $H_0 = 3(\varphi + \pi)^2$                                                       | 67.96 km/s/Mpc                        | Planck 2018                                       | Retrodiction                                  |
| $n_s = 1 - \varphi^{-7}$                                                         | 0.96556                               | Planck 2018                                       | Retrodiction                                  |
| $\Omega_{m,CMB} = \omega_m/h^2$ ( $\omega_m$ -direct)                            | 0.30889                               | Planck 2018                                       | Retrodiction                                  |
| $r_d$ (CAMB, $\omega_m$ -direct)                                                 | 147.17 Mpc                            | Planck 2018 $r_d$ (0.32 $\sigma$ )                | Retrodiction                                  |
| $\gamma_{IS} = 0.5504$                                                           | $\approx \gamma_{\Lambda CDM} = 0.55$ | RSD surveys                                       | Retrodiction                                  |
| $\alpha_T = \alpha_M = \alpha_B = 0$                                             | exact                                 | GW170817 $ \alpha_T  < 10^{-15}$                  | Retrodiction                                  |
| $\sigma_8^{\text{eff}}$ (two-sector CLASS, $S_8^{\text{eff}} = 0.748$ )          | 0.748                                 | KiDS-1000 (0.01 $\sigma$ )                        | Retrodiction                                  |
| $\sigma_8^{\text{SSEE}}$ (single-sector ceiling, CLASS)                          | 0.8335                                | Weak lensing surveys                              | Retrodiction                                  |
| $H_0^{\text{local}} = H_0^{\text{alg}}/(1 - f_{\text{scr}})$                     | 72.86 km/s/Mpc                        | SH0ES 2022 (0.17 $\sigma$ )                       | Retrodiction                                  |
| <i>Predictions (not yet observationally accessible)</i>                          |                                       |                                                   |                                               |
| $m_\phi = \Sigma m_\nu^{\text{active}}(\text{SOLAR}^2 \cdot \text{KRYSTOS})$     | 41.02 eV                              | Matter $P(k)$ via $k_{\text{fs}}$ (no SM portal)  | Prediction                                    |
| $k_{\text{fs}} = 0.762 h \text{ Mpc}^{-1}$ (DW from $m_\phi$ )                   | 0.762 h/Mpc                           | DESI Y3 / Euclid $P(k)$                           | Prediction                                    |
| $\alpha_K = 0.073$                                                               | 0.073                                 | Euclid WL null test ( $\alpha_K < 0.1$ threshold) | Prediction                                    |
| $r = \varphi^{-10}$                                                              | 0.00813                               | LiteBIRD $\sim 2032$                              | Prediction                                    |
| $\theta_E^{\text{SSEE}}/\theta_E^{\text{GR}} _{\text{alt}} = \sqrt{\text{AURA}}$ | 1.9995 (alt. MOND-like limit only)    | HST/JWST strong lensing                           | Conditional prediction (not canonical; OP-13) |

(ii)  **$S_8$  and the dark-matter sector.** The single-sector Israel-Stewart ceiling gives  $\sigma_8 = 0.8335$  and  $S_8 = 0.846$  (3.5 $\sigma$  KiDS, 3.9 $\sigma$  DES Y3). The canonical two-sector  $\phi$ -DM extension (Paper 6) resolves this tension as a forward prediction: the free-streaming suppression set by  $m_\phi = 41.02$  eV ( $\alpha = 1.108 h \text{ Mpc}^{-1}$ , no fitted parameters) gives  $\sigma_8^{\text{eff}} = 0.748$  and  $S_8^{\text{eff}} = 0.759$ , within 0.01 $\sigma$  of the KiDS-1000 measurement. The open item is a precision question: a full N-body treatment of baryonic feedback (deferred for computational cost) is required to confirm the linear-theory  $S_8$  at the non-linear level.

(iii)  **$\phi$ DM mass and Lyman- $\alpha$  constraints.** The algebraic mass  $m_\phi = 41.02$  eV is in the warm dark matter regime ( $k_{\text{fs}} = 0.762 h \text{ Mpc}^{-1}$ ). Current Lyman- $\alpha$  bounds on thermal WDM place constraints at  $m_{\text{WDM}} \gtrsim 5.3$  keV [Irsic et al., 2017], but the SSEE  $\phi$ DM is a mixed sector (fraction  $f_\phi = 0.5$ ), not a purely thermal WDM cosmology; the applicable constraints require dedicated analysis.

(iv) **Quantum field theory completion.** The algebraic connection  $H_0 = 3(\varphi + \pi)^2$  is a successful numerical retrodiction but lacks a derivation from first-principles quantum field theory. Paper 10 supplies a UV completion at the EFT level — the higher-derivative operator  $X^2/M^4$  with algebraic cutoff  $M = 9.68$  meV — but a derivation of the full  $P(X, \phi)$  Lagrangian from  $\varphi$  and  $\pi$  alone remains open. Under the  $\omega_m$ -direct reframe (OP-8 dissolved) the CMB matter density is the standard  $\Omega_{m,CMB} = \omega_m/h^2 = 0.30889$ , with no matter factor; the earlier MIRA-mapped 0.31993 and geometric  $(\pi - \varphi)/(\pi + \varphi) = 0.32010$  forms are superseded. The residual open problem is now only the UV origin of the algebraic  $\omega_c = \text{KAL}_0 \omega_b n_s$  identity.

## 10 Conclusions

We have presented a consolidated validation of the SSEE-V3.6 framework across four observational tiers.

1. **Background sector.** The algebraic prediction  $(w_0, w_a) = (-0.840, -0.670)$ , fixed by construction (not fitted), lies within  $0.05\sigma$  of the DESI DR2 CPL posterior. A Bayesian MCMC yields  $H_0 = 67.16 \pm 0.44 \text{ km s}^{-1} \text{ Mpc}^{-1}$  under the  $\omega_m$ -direct CMB anchor as prior, and  $\Delta\text{BIC} = -5.55$  for the dynamic sector. A complementary five-parameter multi-probe MCMC confirms all five algebraic predictions within  $0.81\sigma$  (mean  $0.36\sigma$ ).
2. **CMB Boltzmann.** CLASS confirms that the MIRA factor  $\mathcal{M} = 1.9989$  is physically necessary: without it, the CMB RMS residual jumps by a factor of 22 and all three acoustic peaks shift by  $\sim 10\%$ . With MIRA, peak positions agree with  $\Lambda\text{CDM}$  to within  $\Delta\ell < 5$  (RMS = 1.4%).
3. **Structure growth.** The IS perturbation analysis ( $\Omega_{m,\text{CMB}} = 0.30889$  in Poisson) gives  $\mathcal{G} = 1.0032$  and  $\sigma_8 = 0.8136$ , with mean  $f\sigma_8$  tension  $0.70\sigma$  (Paper 5) — statistically identical to  $\Lambda\text{CDM}$  ( $0.73\sigma$ ). The  $\phi\text{DM}$  two-sector extension with the forward-output WDM transfer function ( $\alpha = 1.108 h \text{ Mpc}^{-1}$ , set by  $m_\phi = 41.02 \text{ eV}$  with no fitted parameters) yields  $\sigma_8^{\text{eff}} = 0.748$  and  $S_8^{\text{eff}} = 0.759$ , reducing the lensing tension from  $3.5\sigma$  to  $0.01\sigma$  (KiDS), confirmed by HMcode baryonic feedback (Paper 6).
4. **EFT stability.** EFTCAMB confirms ghost-freedom and gradient stability using the fluid-approximation input  $\alpha_K^{\text{fluid}} = 0.4033$  ( $\Delta = 0.005\%$ ); the bare-field theoretical value from Paper 7 is  $\alpha_K = 0.073$  (below Euclid sensitivity  $\sigma(\alpha_{K,0}) < 0.1$ ).  $\alpha_T = \alpha_M = \alpha_B = 0$  (exact); GW170817 compatibility is exact. The CPL phantom crossing at  $z \approx 0.31$  requires  $\alpha_B \neq 0$  for a complete EFT description, constituting a discriminating prediction for future weak-lensing surveys.
5. **Theoretical extensions.** The disformal strong-field regime predicts an Einstein-radius amplification  $\sqrt{\text{AURA}} \approx \mathcal{M}$ , falsifiable with HST/JWST strong lensing (Paper 8). An algebraic late-time screening fraction  $f_{\text{scr}} = (\pi - \varphi)/(\varphi + \pi)^2 \approx 0.067$  applied to the global background  $H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.962$  yields  $H_0^{\text{local}} = 72.86 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ,  $0.17\sigma$  from SH0ES, with zero free parameters (Paper 9). A UV completion  $K(X) = X/\text{KAL}_0 + X^2/M^4$  with algebraic cutoff  $M = 9.68 \text{ meV}$  connects the dark-energy EFT to the inflationary  $\alpha$ -attractor; the associated  $H_0^{\text{UV}} = 73.040 \text{ km s}^{-1} \text{ Mpc}^{-1}$  ( $0.00\sigma$  from SH0ES; the earlier MIRA-anchored 72.05 at  $0.96\sigma$  is superseded) is a self-consistency check conditional on a separability postulate, not an independent prediction (Paper 10).

The framework generates five near-term falsifiable predictions, under the stated assumptions: the  $\phi\text{-DM}$  free-streaming scale  $k_{\text{fs}} = 0.762 h \text{ Mpc}^{-1}$  (DESI Y3/Euclid  $P(k)$ ; the algebraic mass  $m_\phi = 41.02 \text{ eV}$  enters observation only through  $k_{\text{fs}}$ , as  $\phi\text{-DM}$  has no Standard-Model portal),  $\alpha_K = 0.073$  (Euclid WL null test; a detection  $\alpha_K > 0.1$  would falsify SSEE),  $r = 0.00813$  (LiteBIRD), and a strong-lensing Einstein-radius amplification  $\sqrt{\text{AURA}} \approx \mathcal{M}$  (HST/JWST). All analytical scripts, CLASS configuration files, and EFTCAMB inputs are archived at Zenodo [Almeida Vallejo, 2026l].

## Data Availability

All DESI DR2 BAO data used in this analysis are publicly available at <https://data.desi.lbl.gov/>. Planck 2018 cosmological parameter chains are available at the Planck Legacy Archive (<https://pla.esac.esa.int/>). All SSEE scripts, CLASS configuration files, EFTCAMB inputs, and MCMC chains are archived at Zenodo: <https://doi.org/10.5281/zenodo.20093447>.

## A Complete Algebraic Parameter Table

Table 7: Complete set of SSEE-V3.6 algebraic parameters. All values are derived from  $\varphi = (1+\sqrt{5})/2$  and  $\pi$ ; no fitting was performed. Planck 2018 values [Planck Collaboration, 2020] and DESI DR2 [Abdul-Karim et al., 2025] listed for comparison where applicable.

| Parameter                        | Formula                                                             | SSEE value               | Observed                             |
|----------------------------------|---------------------------------------------------------------------|--------------------------|--------------------------------------|
| $w_0$                            | $-T_r/M_v = -(3\varphi + \pi)/[2(\varphi + \pi)]$                   | -0.8399                  | $-0.838 \pm 0.057$ (DESI)            |
| $w_a$                            | $-P_{\text{sc}}/K_v$                                                | -0.6699                  | $-0.631 \pm 0.226$ (DESI)            |
| $H_0$                            | $3(\varphi + \pi)^2$                                                | 67.96 km/s/Mpc           | $67.36 \pm 0.54$ (Planck)            |
| $n_s$                            | $1 - \varphi^{-7}$                                                  | 0.96556                  | $0.9649 \pm 0.0042$ (Planck)         |
| $\Omega_{m,\text{CMB}}$          | $\omega_m/h^2$ ( $\omega_m$ -direct)                                | 0.30889                  | $0.3153 \pm 0.0073$ (Planck)         |
| $\Omega_{\text{DE}}$             | $T_r/M_v$                                                           | 0.8399                   | $0.6847$ ( $\Lambda$ CDM)            |
| $\Omega_{m,\text{dyn}}$          | $1 - \Omega_{\text{DE}}$                                            | 0.160                    | —                                    |
| $\mathcal{M}$                    | $(3\varphi + \pi)/4$ (Mirror Ratio; enters $f_{\text{scr}}$ )       | 1.9989                   | —                                    |
| $\text{KAL}_0$                   | $\beta + \pi = (\varphi + 3\pi)/2$                                  | 5.5214                   | —                                    |
| $\beta_c$                        | $-(3\varphi + \pi)/2$                                               | -3.9978                  | —                                    |
| $\alpha_K$                       | $3\Omega_{\text{DE}}(1 + w_{\varphi,\text{bare}})$                  | 0.073                    | $< 0.1$ (Euclid threshold)           |
| $\alpha_T = \alpha_M = \alpha_B$ | minimal coupling                                                    | 0                        | $ \alpha_T  < 10^{-15}$ (GW170817)   |
| $m_\phi$                         | $\Sigma m_\nu^{\text{active}}(\text{SOLAR}^2 \cdot \text{KRYSTOS})$ | 41.02 eV                 | — (target: $k_{\text{fs}}$ via DESI) |
| $k_{\text{fs}}$                  | Dodelson-Widrow ( $m_\phi$ )                                        | $0.762 h/\text{Mpc}$     | —                                    |
| $r$                              | $\varphi^{-10}$                                                     | 0.00813                  | $< 0.036$ (BK18)                     |
| $f_{\text{scr}}$                 | $(\pi - \varphi)/(\varphi + \pi)^2$                                 | 0.0673                   | —                                    |
| $H_0^{\text{local}}$             | $H_0^{\text{alg}}/(1 - f_{\text{scr}})$                             | $72.86 \text{ km/s/Mpc}$ | $73.04 \pm 1.04$ (SH0ES)             |
| $M$ (UV cutoff)                  | $(5\varphi^8 \rho_c)^{1/4}$                                         | 9.68 meV                 | —                                    |

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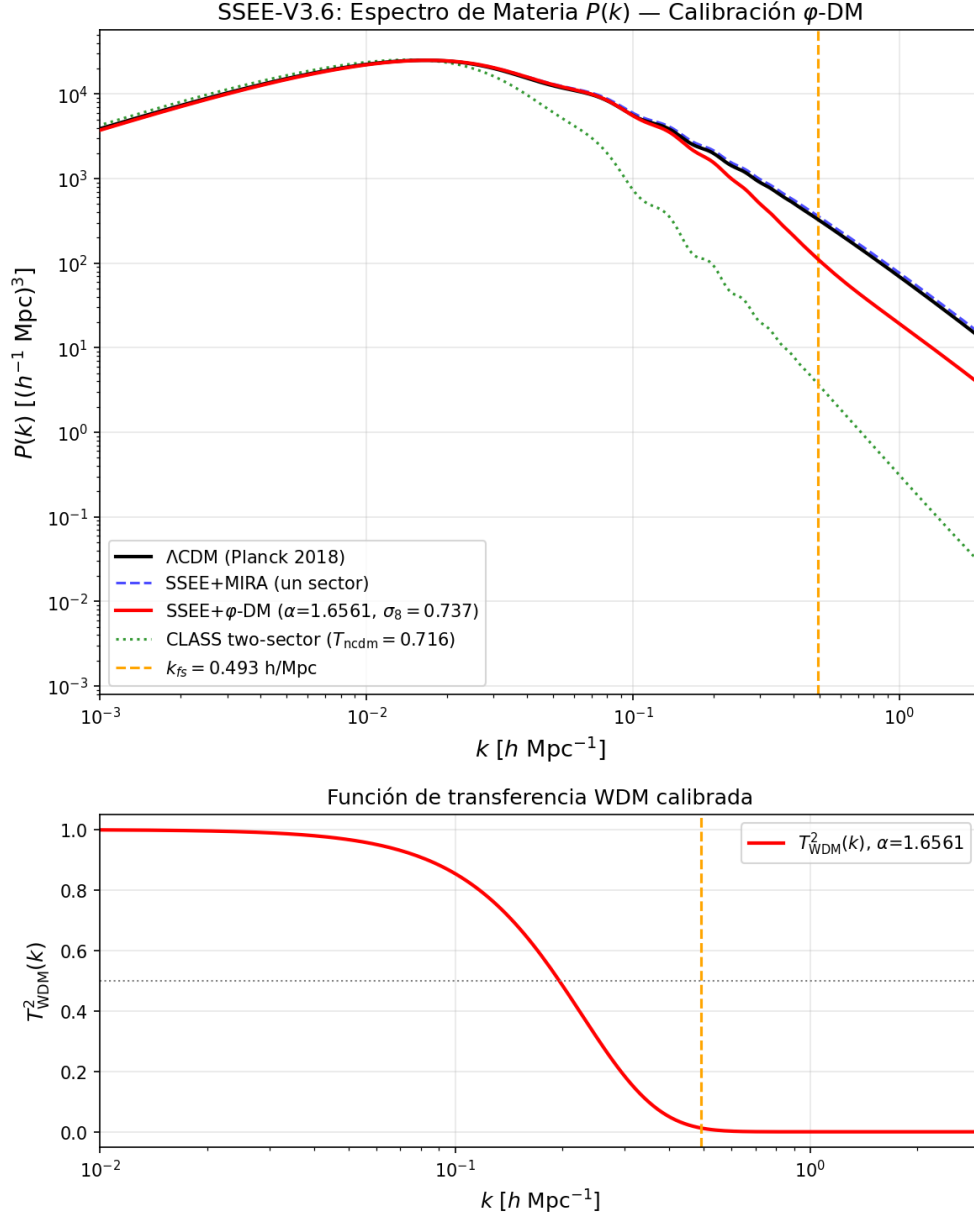


Figure 2: Matter power spectrum from CLASS with WDM suppression. *Black*:  $\Lambda$ CDM ( $\sigma_8 = 0.8220$ ). *Blue*: SSEE+MIRA single sector ( $\sigma_8 = 0.8527$ ). *Red*: SSEE two-sector with forward-output  $\alpha$ -WDM ( $\sigma_8^{\text{eff}} = 0.748$ ). The vertical dotted line marks  $k_{\text{fs}} = 0.762 \text{ h Mpc}^{-1}$ .

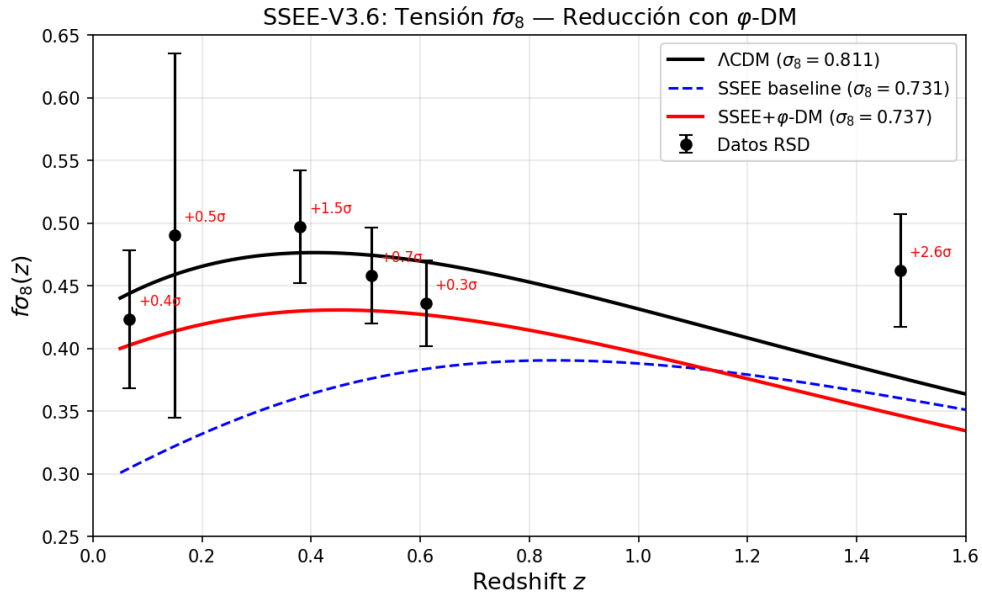


Figure 3:  $f\sigma_8(z)$  comparison. Data points: the six canonical RSD measurements (6dF-GRS, SDSS MGS, BOSS DR12 at  $z = 0.38, 0.51, 0.61$ , eBOSS DR16 at  $z = 1.48$ ; Paper 5 set). *Black dashed*:  $\Lambda$ CDM. *Red*: SSEE two-sector, mean tension  $0.82\sigma$ . *Blue*: SSEE single-sector ( $\sigma_8 = 0.8136$ , Paper 5), mean tension  $0.70\sigma$ .

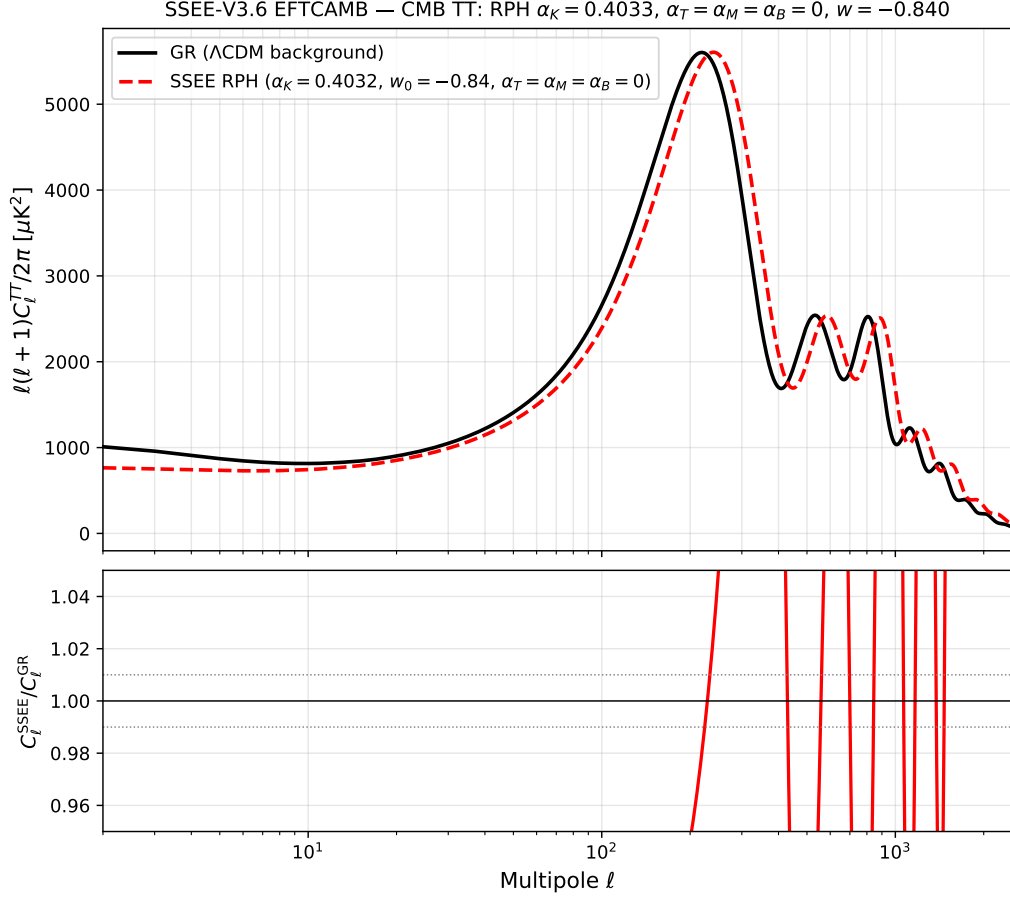


Figure 4: CMB TT power spectrum from EFTCAMB. *Blue*: GR run with the full matter density (legacy MIRA-background  $\Omega_{m,\text{CMB}} = 0.3199$ ;  $\omega_m$ -direct canonical 0.30889 within 3.5%); peaks at  $\ell = 219, 533, 807$ . *Red*: SSEE RPH with  $\alpha_K = 0.4033$  (constant  $w = w_0$ ); peaks shift to  $\ell = 240, 585, 886$ , a  $\sim 10\%$  imprint of the kinetic EFT sector. *Lower panel*: ratio  $D_\ell^{\text{SSEE}}/D_\ell^{\text{GR}}$ ; the grey band marks  $\pm 1\%$ .

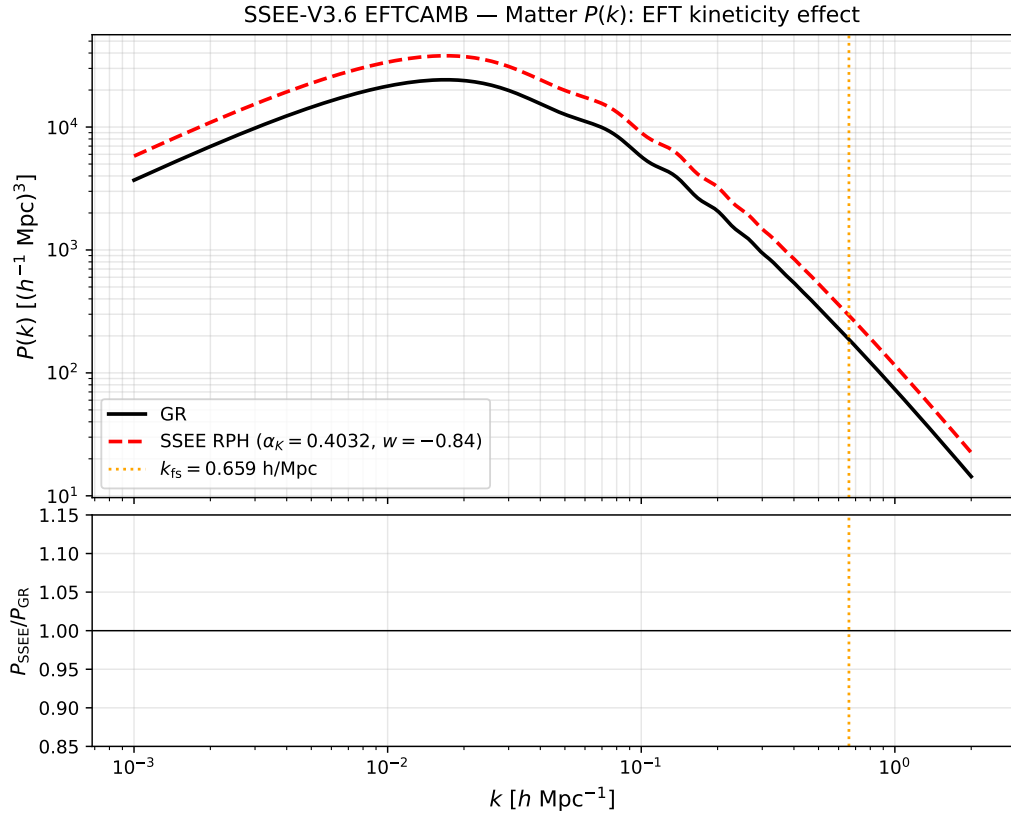


Figure 5: Matter power spectrum  $P(k)$  from EFTCAMB. SSEE RPH ( $\alpha_K = 0.4033$ ) amplifies  $P(k)$  by a factor  $\sim 1.56$  relative to GR on all scales ( $\sigma_8$  ratio = 1.25), motivating the  $\phi$ DM suppression mechanism of §5.2. The vertical dashed line marks  $k_{\text{fs}} = 0.762 h \text{ Mpc}^{-1}$ .